

Is the ESG Dimension Spanned by the Fama–French Factors?

A Complementary-Factor Analysis of Size-Neutral ESG Portfolios in the STOXX Universe

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Abstract

In a companion paper we show that the interaction of Refinitiv ESG scores with retail sentiment does not *predict* next-month STOXX returns once look-ahead bias — both the sort/return misalignment and the ESG publication lag — is removed. This paper asks the question that survives the null: do ESG-sorted portfolios carry a distinct source of *contemporaneous* common return variation that the standard factor model misses? We construct the ESG Sentiment Factor (ESF) exactly as Fama and French construct SMB and HML: an independent 2×3 double sort on size and the (publication-lagged) ESG score, value-weighted, formed at month-end t and held over month $t+1$. Size neutralisation is essential here, because ESG scores and coverage rise mechanically with firm size. Over 154 months of sufficient ESG coverage (2013–2025), ESF earns -0.24% per month. The central result is the spanning regression: against the European FF5 + momentum factors, ESF leaves an unspanned alpha of -0.47% per month ($t = -2.21$, $p = 0.027$) with $R^2 = 0.06$ and no pairwise factor correlation above 0.16. The Fama–French factors do not span the ESG dimension — and the unspanned average return is *negative*: a size-neutral European “greenium” in which high-ESG stocks earn systematically less than low-ESG peers net of all six factor exposures, consistent with [Pástor et al. \(2021\)](#). Adding ESF as a seventh right-hand-side factor leaves the 25 ESG/SAT test-asset alphas essentially unchanged, so the factor is not repackaged test-asset mispricing, while the test assets’ ESF loadings line up monotonically with their ESG sort. We conclude with the practical implication: whether or not ESG characteristics forecast returns, European performance attribution and anomaly evaluation should test exposures against an ESG factor on the right-hand side — the same role SMB and HML play — rather than treat ESG as a forecasting signal.

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1 Introduction

A factor does not need to forecast returns to matter for asset pricing. The Fama–French factors are not return forecasters; they are contemporaneous return spreads whose role is to absorb common variation and reprice test assets. The economically meaningful question about a candidate factor is therefore not “does it predict?” but “is it spanned?”: can its average return and variance be reproduced by a portfolio of the existing factors (Barillas and Shanken, 2017; Fama and French, 2018)?

This paper asks that question for the ESG dimension of European equities. The companion paper establishes the predictive null: ESG scores, alone or interacted with retail sentiment, do not forecast next-month STOXX returns once formation/holding alignment is enforced and scores are lagged to their plausible publication dates. Yet the theoretical literature gives strong reasons to expect ESG to define a *factor* in the covariance sense. Pástor et al. (2021) model green assets as carrying a distinct ESG-taste shock: when sustainability preferences or regulation shift unexpectedly, green stocks move together against brown stocks, generating common variation orthogonal to cash-flow fundamentals. Pedersen et al. (2021) reach the same structure from an ESG-information channel. The post-2018 European regulatory wave (Action Plan, taxonomy, SFDR) is exactly the kind of repeated, market-wide taste shock that should make a green-minus-brown portfolio a covarying return series rather than noise.

We test this directly with the Fama–French 2×3 construction: at each month-end, stocks are independently split by size (median) and by the six-month-lagged Refinitiv ESG score (30th/70th percentiles); six value-weighted portfolios are held over the following month; ESF is the average high-ESG return minus the average low-ESG return, neutralised across size legs. Size neutralisation does real work for ESG — large firms disclose more and score systematically higher (Berg et al., 2021) — so an unneutralised spread would be substantially a (short) size bet.

Three results follow. First, the unconditional ESF premium is *negative*: -0.24% per month (annualised Sharpe -0.36) over the 154 months with sufficient coverage. Second — the paper’s central finding — ESF is *not spanned* by the European FF5 + momentum factors: the spanning regression leaves an alpha of -0.47% per month ($t = -2.21$) with an R^2 of only 0.06, and ESF’s largest absolute correlation with any FF factor is 0.16. The negative sign makes the result a size-neutral greenium: investors in high-ESG European names have earned less than in comparable low-ESG names, net of every standard factor exposure — the pattern equilibrium models of ESG tastes predict. Third, adding ESF as a seventh factor to regressions of the 25 ESG/SAT test portfolios leaves their alphas essentially unchanged while the ESF loadings line up monotonically with the ESG sort — the covariance signature of a genuine common factor rather than a repackaging of the test assets.

2 Why Spanning, Not Prediction

A *predictive signal* is information at t that forecasts returns over $t+1$; the relevant tests are sorted-portfolio spreads and Fama–MacBeth slopes on lagged characteristics. A *pricing factor* is a contemporaneous return series that co-moves with test assets; the relevant tests are spanning regressions among factors and test-asset alpha reductions (Barillas and Shanken, 2017). The companion paper shows the ESG dimension fails the first test in Europe. That is not disqualifying — most accepted factors would fail it too. The question here is the second test.

For ESG specifically, the spanning framing also sidesteps the least productive part of the “does ESG pay?” debate. Pástor et al. (2021) imply that realised green out-performance can coexist with lower green *expected* returns whenever tastes strengthen unexpectedly — so realised-return horse races are uninformative about premia. Whether the factor is spanned, by contrast, is a well-posed covariance question that realised data can answer.

Remark 1. Throughout, *ESF* denotes the traded long–short return series of Definition 1, not the ESG characteristic itself.

3 Data

The stock panel is identical to the companion paper: STOXX Europe 600 constituents, June 2009 – December 2025, financials excluded, Refinitiv EUR total returns, volume, market capitalisation, and TR.TRESGScore. The ESG score is lagged six months to respect publication delay, treated as missing when exactly zero, and winsorised at 1%/99% within month. European FF5 + momentum factors come from the Ken French Data Library. ESG coverage grows over the sample; months where any 2×3 corner leg holds fewer than five stocks are dropped from the factor series.

4 Factor Construction

Definition 1 (ESF factor). *At the end of each formation month t , all eligible stocks are independently sorted by (i) market capitalisation, split at the cross-sectional median into Small (S) and Big (B), and (ii) the lagged ESG score, split at the 30th and 70th percentiles into Low (L), Mid, and High (H). The four corner portfolios are value-weighted with month- t weights and held over month $t+1$:*

$$\text{ESF}_{t+1} = \frac{1}{2}(r_{t+1}^{SH} + r_{t+1}^{BH}) - \frac{1}{2}(r_{t+1}^{SL} + r_{t+1}^{BL}). \quad (1)$$

The same sort-at- t / hold-at- $t+1$ discipline as the companion paper applies: forward

returns are taken only when the stock’s next observation is exactly one month ahead, and automated tests verify that a panel in which the ESG score drives only formation-month returns produces an identically zero factor.

Table 1: ESF Factor: Summary Statistics

Statistic	Value
Mean (%/month)	-0.243
Std. dev. (%/month)	2.31
Annualised Sharpe ratio	-0.36
Skewness	-0.78
Excess kurtosis	5.09
Months	154

The premium is negative — -0.24% per month with a monthly standard deviation of 2.31% (annualised Sharpe -0.36). The 154-month span reflects ESG coverage: the factor series starts once every 2×3 corner leg reliably holds at least five stocks, which excludes the thin-coverage years before 2013. A negative mean is not a defect of the factor; it is the finding. In the [Pástor et al. \(2021\)](#) equilibrium, green assets earn lower expected returns precisely because investors derive non-pecuniary utility from holding them and because green assets hedge climate-taste shocks.

5 Spanning Results

5.1 Correlations

Table 2: Pairwise Correlations of ESF with the FF5 + Momentum Factors

FF factor	Corr. with ESF
Mkt-RF	+0.157
SMB	+0.027
HML	+0.062
RMW	-0.023
CMA	-0.034
UMD	+0.042

No correlation exceeds 0.16 in absolute value, and every non-market correlation is below 0.07. The near-orthogonality to SMB is worth emphasising: raw ESG scores correlate

strongly with firm size (Berg et al., 2021), so an unneutralised green-minus-brown spread would load heavily on size. The 2×3 construction removes essentially all of it. Whatever variation ESF captures, it is not a linear repackaging of an existing factor.

5.2 The Spanning Regression

$$\text{ESF}_t = \alpha + \beta_1 \text{MktRF}_t + \beta_2 \text{SMB}_t + \beta_3 \text{HML}_t + \beta_4 \text{RMW}_t + \beta_5 \text{CMA}_t + \beta_6 \text{UMD}_t + \varepsilon_t. \quad (2)$$

Under the null that the FF factors span ESF, $\alpha = 0$: a mean–variance investor already holding the six factors gains nothing from adding ESF. A significant α means ESF expands the frontier of the factor set (Barillas and Shanken, 2017; Fama and French, 2018).

Table 3: Spanning Regression: ESF on the European FF5 + Momentum Factors

Variable	Coefficient	<i>t</i> -stat	<i>p</i> -value
α	-0.473%**	-2.21	0.0273
Mkt-RF	0.103**	2.37	0.0176
SMB	-0.030	-0.28	0.7768
HML	0.182	1.41	0.1584
RMW	0.039	0.21	0.8309
CMA	-0.187	-0.90	0.3696
UMD	0.159**	2.52	0.0119
R^2	0.058		
N (months)	154		

Notes: Monthly observations; Newey–West standard errors with 6 lags. The alpha is expressed in percent per month. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The unspanned alpha is -0.47% per month ($t = -2.21$, $p = 0.027$), almost double the raw mean in magnitude: ESF carries small positive loadings on the market ($\beta = 0.10$, $t = 2.37$) and momentum ($\beta = 0.16$, $t = 2.52$) which contributed positively to its realised return, so netting them out deepens the shortfall. The R^2 of 0.06 says the six factors jointly explain six percent of ESF’s variance: spanning fails comprehensively, not marginally. In the Barillas and Shanken (2017) framework the sign of the alpha is irrelevant to the verdict — a significantly negative alpha expands the mean–variance frontier of the factor set exactly as a positive one does (via the short side). Economically, however, the sign is the interesting part: over a sample that *contains* the 2019–2021 green repricing wave — the period in which taste shocks should have flattered green returns — the size-neutral European greenium is still negative net of FF6. This strengthens rather than weakens the equilibrium reading: if realised returns understate the expected-return gap during taste-strengthening periods (Pástor et al., 2021), the steady-state greenium is, if anything, more

negative than the estimate.

5.3 Does ESF Explain the Test Assets?

Table 4: Test-Asset Alphas: FF6 versus FF6 + ESF (25 ESG/SAT Portfolios)

Model	Mean $ \alpha $ (%/month)	$\# t > 2$	Mean R^2
FF6	0.562	6	0.333
FF6+ESF	0.553	5	0.371

Table 5: ESF Loadings of the 25 ESG/SAT Portfolios (FF6 + ESF Model)

ESG Quintile	SAT Quintile				
	Q1 (Bear)	Q2	Q3	Q4	Q5 (Bull)
Q1 (Low ESG)	-0.37	-0.36	-0.29	-0.34	-0.10
Q2	-0.20	0.10	-1.14	-0.10	-0.33
Q3	0.01	0.09	-0.20	0.04	0.08
Q4	-0.17	0.01	0.15	0.69	0.01
Q5 (High ESG)	0.44	0.31	0.95	0.24	0.42

Notes: Slope coefficients on ESF from time-series regressions of each portfolio’s excess return on the six FF factors plus ESF.

Two observations. First, mean absolute test-asset alphas are essentially unchanged when ESF is added (0.562% to 0.553% per month; significant count 6 to 5): ESF is not manufacturing its spanning alpha by absorbing test-asset mispricing, consistent with the companion paper’s finding that these portfolios contain no sentiment-related alpha after correct alignment. Mean R^2 rises from 0.33 to 0.37 — the factor adds explanatory power for common variation, which is its actual job. Second, the ESF loadings in Table 5 rise monotonically with the ESG sort: the low-ESG row loads around -0.3 in every SAT column and the high-ESG row around $+0.3$ to $+0.9$. The factor covaries with the characteristic that defines it, across an independent second sort dimension — the cross-sectional signature ? require of a genuine common factor.

6 Interpretation and Caveats

6.1 What ESF Is

The results support a specific claim: ESG-sorted, size-neutral European portfolios carry a source of common return variation that is close to orthogonal to the FF5 + momentum block, prices at a significantly negative premium, and cannot be reproduced by the incumbent factors. The natural applications are on the right-hand side:

1. **Performance attribution.** A European fund tilted green will show a negative FF6 alpha that is in fact ESF exposure — and a brown-tilted fund a positive one. Adding ESF separates stock selection from the sustainability tilt.
2. **Anomaly evaluation.** New European anomalies should demonstrate robustness to ESF, precisely because ESG comovement is unspanned by the standard block.
3. **Risk decomposition.** ESF gives risk models a measurable exposure to climate-taste and ESG-regulation shocks, which the post-2018 policy wave delivers repeatedly.

6.2 What ESF Is Not

ESF is not a timing signal: the companion paper’s predictive null is unambiguous. Nor can we say whether the negative premium is a taste-based equilibrium discount, compensation for hedging value, or slow repricing — 154 months cannot separate these; what they can establish is that the premium is not a disguised exposure to any existing factor.

Three caveats bound any conclusion. First, per [Harvey et al. \(2016\)](#), new factor claims should clear $t \approx 3.0$; results below that threshold make ESF a candidate, not an established factor. Second, the sample is one universe (STOXX) and one score vendor (Refinitiv); ESG ratings disagree notoriously across vendors ([Berg et al., 2021](#)), so vendor robustness is the obvious next step. Third, the regulatory-taste-shock mechanism implies the factor’s behaviour may be regime-dependent, and the post-SFDR subsample is still short.

7 Conclusion

The ESG dimension of European equities does not forecast returns, but it is not redundant either. Constructed with the standard Fama–French 2×3 methodology, strict formation/holding alignment, and a publication-realistic score lag, the ESG Sentiment Factor earns -0.24% per month, is nearly uncorrelated with every FF5 + momentum factor, and leaves a spanning alpha of -0.47% per month ($t = -2.21$) against the full

six-factor block with an R^2 of 0.06. The Fama–French factors do not span the ESG dimension, and the unspanned premium is a negative, size-neutral European greenium — the sign equilibrium theory predicts when investors hold green assets partly for non-pecuniary reasons. The constructive conclusion mirrors the RSF sibling: treat ESF as a contemporaneous regressor that strips a distinct dimension of common variation out of returns — something to control for, and, for a brown-tilted investor, something whose premium exists to be harvested only with eyes open to its taste-based and regulatory nature.

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